

Natural Language Processing

Introduction to NLP and Linguistics

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Course Overview

- **Topics Covered:**

- ☐ Introduction to NLP
- ☐ Text Preprocessing
- ☐ Text Representation
- ☐ Word Embeddings
- ☐ Language Models
- ☐ RNNs
- ☐ Transformers
- ☐ Text Classification
- ☐ Named Entity Recognition
- ☐ Large Language Models

What is Language?

- Language is a communication system used by humans to express ideas, emotions, and information.
- Language consists of:
 - Sounds
 - Words
 - Grammar
 - Meaning
 - Context

Natural Language

- Natural language refers to languages used by humans for communication.
- Examples:
 - Arabic
 - English
 - French
 - Chinese
- Note:

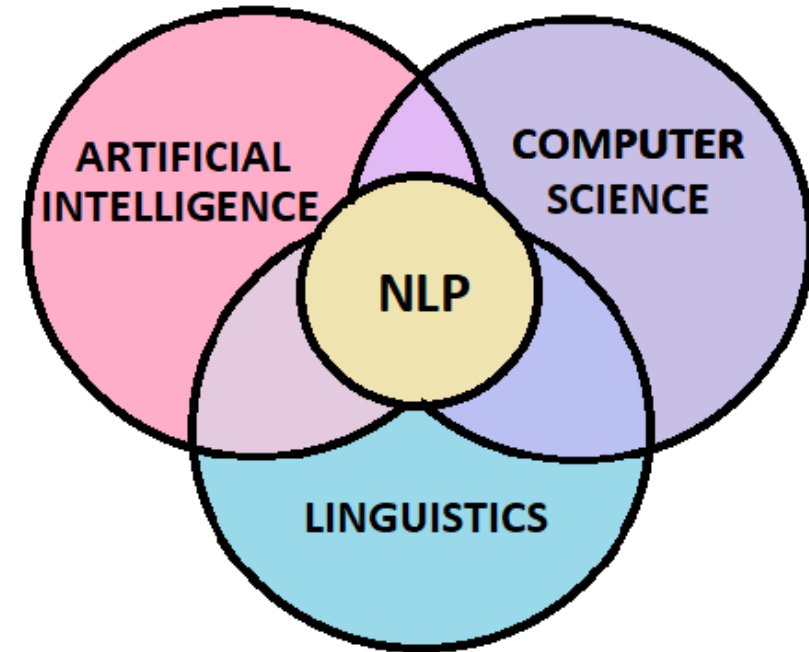
Natural languages evolve over time and contain ambiguity and complexity.

Programming Languages vs Natural Languages

Programming Language	Natural Language
Python	Arabic
Java	English
Machine instructions	Human communication
Precise	Ambiguous

What is NLP?

- Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, analyze, process, and generate human language.
- NLP combines:
 - Linguistics
 - Computer Science
 - Artificial Intelligence



Goals of NLP

- The major goals of NLP include:
 - Understanding language
 - Generating language
 - Extracting information
 - Answering questions
 - Translating languages
 - Building conversational systems



Why Do We Study NLP?

- Large amounts of information are available in textual form.
- NLP enables machines to process and utilize this information effectively
- Benefits:
 - Automating language tasks
 - Improving human-computer interaction
 - Extracting knowledge from text
 - Supporting intelligent applications

Importance of NLP

- NLP has become essential in modern AI applications.
- Examples include:
 - Search Engines
 - Digital Assistants
 - Social Media Analysis
 - Recommendation Systems
 - Customer Support

Applications of NLP

- Natural Language Processing has numerous applications in real-world systems.
- Major applications include:
 - Machine Translation
 - Sentiment Analysis
 - Question Answering
 - Information Retrieval
 - Text Summarization
 - Speech Recognition
 - Named Entity Recognition
 - Chatbots

Machine Translation

- Machine Translation automatically converts text from one language into another.
- Applications:
 - Google Translate

Sentiment Analysis

- Sentiment Analysis identifies opinions and emotions expressed in text.
- Categories:
 - Positive
 - Negative
 - Neutral
- Examples:
 - 1- "I love this product."
 - Positive
 - 2- "This movie is terrible."
 - Negative

- Applications:
 - Product Reviews
 - Customer Feedback
 - Social Media Analysis

Question Answering

- Question Answering systems provide direct responses to user questions.
- Example
 - Question:
 - What is the capital of France?
 - Answer:
 - Paris
- Applications:
 - ChatGPT
 - Search Engines
 - Educational Systems

Information Retrieval

- Information Retrieval focuses on finding relevant documents from large collections of text.
- Applications:
 - Web Search
 - Search Engines
 - Document Retrieval

Text Summarization

- Text Summarization generates shorter versions of long documents while preserving important information.
- Applications:
 - News Articles
 - Research Papers
 - Reports

Speech Recognition

- Speech Recognition converts spoken language into text.
- Applications:
 - Voice Assistants
 - Subtitles
 - Speech-to-Text Systems



Named Entity Recognition (NER)

- Named Entity Recognition identifies important entities within text.
- Examples:
 - Ahmed → Person
 - Google → Organization
 - Sana'a → Location
 - 2026 → Date
- Applications:
 - News Analysis
 - Healthcare
 - Information Extraction

Chatbots

- Chatbots are systems capable of interacting with users using natural language.
- Examples:
 - ChatGPT
 - Alexa
 - Customer Support Bots
 - Virtual Assistants

History of NLP

- NLP has evolved through several stages:
 - Rule-Based Systems
 - Statistical NLP
 - Machine Learning
 - Deep Learning
 - Transformers
 - Large Language Models

Rule-Based Systems

- Rule-based systems rely on manually designed grammar rules.
- Advantages
 - Easy to interpret
- Disadvantages
 - Difficult to maintain
 - Poor scalability

Statistical NLP

- Statistical NLP uses probabilities and data-driven approaches.
- Examples
 - N-Grams
 - Naive Bayes
 - HMM
 - CRF
- Advantages
 - Learns from data

Deep Learning

- Deep learning introduced neural networks into NLP.
- Examples
 - RNN
 - LSTM
 - GRU
 - Embeddings

Transformers and LLMs

- Transformers revolutionized NLP.
- Examples
 - BERT
 - GPT
 - Gemini
 - DeepSeek

NLP Tasks

- Common NLP Tasks include:
 - Text Classification
 - Part-of-Speech Tagging
 - Named Entity Recognition
 - Translation
 - Summarization
 - Language Modeling
 - Question Answering

Challenges in NLP

- Natural language is difficult for computers.
- Major challenges include:
 - Ambiguity
 - Context Dependence
 - Language Diversity
 - Data Sparsity
 - Low-resource Languages

Ambiguity

- Words and sentences can have multiple meanings.
- Example
 - I saw the man with the telescope.
- Possible meanings:
 - I used a telescope.
 - The man had a telescope.

Context Dependence

- Meaning often depends on context.
- Example:
 - "It is cold here."
- Possible meanings:
 - Information
 - Request
 - Complaint
 - Suggestion

Language Diversity

- More than 7000 languages exist worldwide.
- Languages differ in:
 - Vocabulary
 - Grammar
 - Writing Systems
 - Word Order

Data Sparsity

- Data sparsity occurs when many words or expressions appear very infrequently in the training data.
- Examples
 - Common words
 - the
 - is
 - and
 - Rare words
 - neuro-symbolic
 - bioinformatics
 - cryptoeconomics

Linguistics and NLP

- Linguistics studies language scientifically.
- Important subfields include:
 - Morphology
 - Syntax
 - Semantics
 - Pragmatics

Morphology

- Morphology studies word formation.
- Examples:
 - play
 - playing
 - played
 - player

Syntax

- Syntax studies sentence structure.
- Example:
 - The boy eats an apple.
 - Correct
 - Apple eats boy.
 - Incorrect

Semantics

- Semantics studies meaning.
- Example:
 - Bank
 - Financial Institution
 - River Bank

Pragmatics

- Pragmatics studies meaning in context.
- Example
 - "It is hot here."
- Could imply:
 - Information
 - Request
 - Complaint

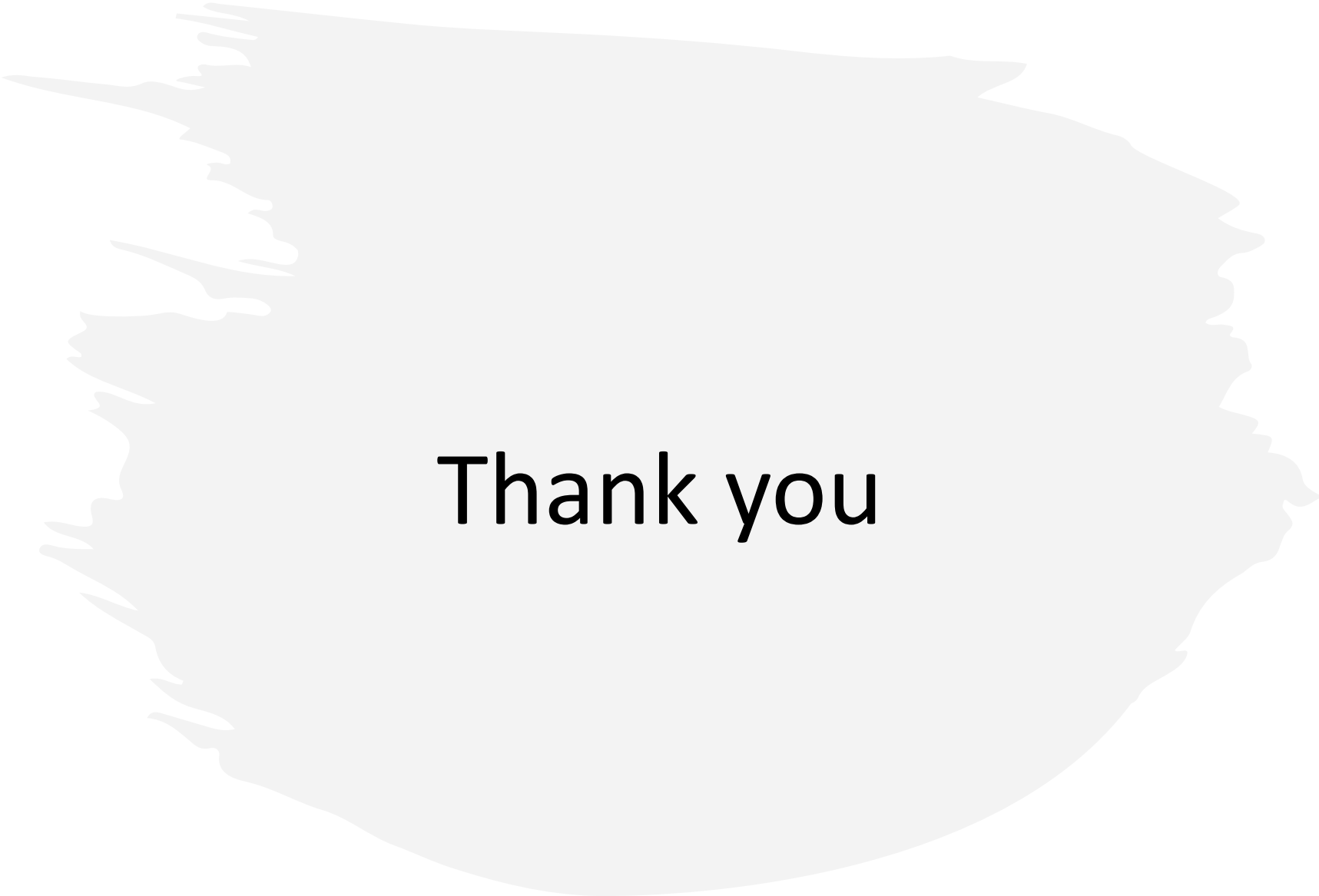


NLP Pipeline

- Raw Text
- ↓
- Cleaning
- ↓
- Tokenization
- ↓
- Representation
- ↓
- Modeling
- ↓
- Evaluation
- ↓
- Prediction

Modern NLP

- Recent technologies include:
 - Embeddings
 - Transformers
 - BERT
 - GPT
 - RAG
 - Agents



Thank you

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